



Systems Programming with Python Coursework: June 2026 (50%)

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INSTRUCTIONS

Identify a real-world problem from any of the project areas listed below and develop a functional Python application that addresses the problem effectively. The system may be desktop-based or web-based, depending on your design choice.

Project Areas (Select One)

- Education Systems
- Healthcare Systems
- Business and Commerce
- Transport and Logistics
- Agriculture and Farming
- Community and Social Services
- Financial Services
- Security and Safety Systems
- E-Government Services
- Hospitality and Tourism
- Real Estate and Property Management
- Retail and Inventory Management
- Event Management Systems
- Library Management Systems
- Human Resource Management Systems
- School Management Systems
- Online Booking and Reservation
- Communication and Messaging Systems
- E-Commerce Platforms
- Environmental and Waste Management

For instance, a local clinic may struggle with keeping patient records manually, leading to misplaced files and delays in service. You could develop a Clinic Management System that captures patient details, stores records, and retrieves them efficiently. Alternatively, a small retail shop may face difficulties tracking stock levels, and you could design an Inventory Management System to automate stock monitoring and sales recording.

System Requirements

a) Core Functionality

- The application must demonstrate key Python programming concepts.
- It should accept user input, process data, and generate meaningful output.

b) Programming Concepts

- Use of variables and data types
- Control structures (if statements, loops)
- Functions and modular programming
- Data structures such as lists and/or dictionaries
- File handling or database connectivity for persistent data storage
- Implementation of classes and objects
- Use of inheritance and encapsulation
- Proper error and exception handling mechanisms

c) Expectations

- Provide a simple user-friendly interface (menu-driven system)
- Include data validation to ensure accuracy of inputs
- Ensure the system is well-documented and properly structured
- Demonstrate real-life applicability and usability

Report Requirements

- Prepare a detailed report that includes:
- Problem description and justification
- Objectives of the system
- System design (e.g., flowcharts, structure, or architecture)
- Implementation details
- Explanation of how the system works (input, processing, output)
- Testing and evaluation of the system

NB: Submit your report in V-class and ensure to make a class presentation

END